Nearest Neighbor Reasoning

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Teach a neighbour classifier as an inspectable reference-free prediction path.

Supply complete scripts and validation records, separate the book example and reserve the new pool for assessment.

Keep one habit: Require neighbour IDs, distances, votes and settings beside the prediction and score.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Compute absolute **distance**s and rank **training neighbour**s

Teach two 45-minute sessions. Use a single numeric feature to make **distance** arithmetic visible, then distinguish **neighbour** selection from voting. Store only **training** rows.

Alphabetical IDs settle equal **distance**s for this classroom method; explain that it is a convention and other implementations can use different tie behavior.

By the end, I can order **training neighbour**s, apply k and keep query labels out of **prediction**.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- Case exposure check: this source example appears in ai-2/22, ai-4/12. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.
- Printed lesson and workbook
- Number line from 0 to 14 cm and five labelled **training** markers
- Optional Python 3 editor with no packages required
- Blank table for **distance**s ordered IDs **vote**s and **prediction**s

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Similarity by Selected Traits. Give an example and a mistake that the idea helps you avoid.

Look for: Represent items with numeric features

Reference example, not the only acceptable wording: Energy 4, group work 1.

Without opening the old lesson, explain one key idea from Sampling and Representation. Give an example and a mistake that the idea helps you avoid.

Look for: Define a question and target population

Reference example, not the only acceptable wording: Population: all Grade 7 students. Only lunch-club members or one friend group is an unsuitable basis for a whole-grade claim.

Without opening the old lesson, explain one key idea from Over**fitting** Intuition. Give an example and a mistake that the idea helps you avoid.

Look for: Compare **training** and validation **evidence** for model capacity

Reference example, not the only acceptable wording: Teacher explanation: Choose the line with validation MAE 3. The curve **training** MAE 0 does not compensate for validation MAE 8.

It stores a root question and leaf **prediction**s learned from **training**. A new example follows the appropriate branch to a stored leaf **prediction**.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Similarity by Selected Traits

Sampling and Representation

Overfitting Intuition

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

A query has nearest books A and B at **distance** 1, with conflicting labels; C is **distance** 2. A program chooses whichever row occurs first. Explain the risk and define a reproducible alternative.

Concept to explain

A new item can be predicted from nearby labeled examples.

Term	Meaning
neighbor	An example close to another under a specified distance measure.
k	In k-nearest neighbours, the number of nearby examples considered in the prediction.

Term	Meaning
vote	A contribution to a choice, such as one neighbour's label in a classification decision.
tie rule	A predefined method for handling candidates with equal scores or conflicting nearest labels.

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach order, selection and voting before scoring

> Supply complete programs and keep the book pool distinct.

Introduce stored training and the full numeric pool

A nearest-**neighbour** classifier predicts using nearby **training** examples rather than following a fitted tree.

This week we store labelled examples, measure how different their feature values are from a new query, and let selected **neighbour**s **vote**.

Nearby means close under our chosen feature and **distance**, not necessarily similar in every real-world way.

Our new fictional dataset has a **distance** reading in centimetres and a separately reviewed lane label. Each row comes from a distinct trip, and validation trips are separate. Only the numeric reading measures similarity.

IDs identify rows and break **distance** ties; they are not numeric features. These invented readings are not a robot safety system.

For this one-dimensional feature, **distance** is abs(**training** reading minus query reading). abs removes the sign. Set k=3 and give each selected **neighbour** one **vote**.

Order equal **distance**s by ID alphabetically, declared before scoring. In this binary, three-**vote** exercise the label **vote** cannot tie, but the **distance**s can.

Trace the single-query program

```
train = [("A", 2, "blocked"), ("B", 4, "clear"),
         ("C", 8, "clear"), ("D", 10, "blocked"),
         ("E", 12, "clear")]
query = 3
ranked = sorted((abs(x - query), item_id, label)
                for item_id, x, label in train)
near = ranked[:3]
labels = [row[2] for row in near]
prediction = "clear"
if labels.count("blocked") > labels.count("clear"):
    prediction = "blocked"
print([row[1] for row in near])
print(prediction)
```

For query 3, **distance**s to A through E are 1,1,5,7,9 cm. sorted orders tuples by **distance** first, then the unique ID. ranked[:3] takes the first three rows.

The **output** is ['A', 'B', 'C'] then clear: one blocked **vote** and two clear **vote**s.

If k were 1 under the same **distance tie rule**, A would win and predict blocked. Changing k can change a **prediction**.

We declared k=3 for this exercise; do not choose k after seeing the desired answer and pretend the method was unchanged.

The code assumes unique IDs, numeric readings, only clear or blocked labels and at least three **training** rows. It does not validate those conditions.

A larger k than the stored row count would not produce the promised number of **vote**s. Keep unresolved review labels under a separate documented policy.

Collect the main k comparison and probability limit

Three selected neighbours for query 3

STORED EXAMPLE	FEATURE DISTANCE	ONE LABEL VOTE
A at 2 cm	abs 2 minus 3 is 1	blocked
B at 4 cm	abs 4 minus 3 is 1	clear
C at 8 cm	abs 8 minus 3 is 5	clear

Select neighbours by feature distance and the declared ID tie rule. Read their training labels to vote only after selecting them.

Store the **training** examples and the method settings. At **prediction** time, compare the query only with stored **training** examples. Do not add the query's reference label or other validation labels to the **neighbour** pool.

No cutoff or tree is fitted here, but retaining **training** examples is still part of the learning method.

> **Why do A and B tie for distance to query 3? Which wins for k=1 under our rule, and why does k=3 predict differently? Does the ID rule establish which label is true?**

A **vote** share of two out of three describes these **neighbour**s. It is not a verified two-thirds probability that the new scene is clear. A **prediction** can be wrong even when the nearest examples agree.

Show the full validation and baseline program

```
train = [("A", 2, "blocked"), ("B", 4, "clear"),
         ("C", 8, "clear"), ("D", 10, "blocked"),
         ("E", 12, "clear")]
def predict(query, k):
    ranked = sorted((abs(x - query), item_id, label)
                    for item_id, x, label in train)
    near = ranked[:k]
    labels = [row[2] for row in near]
    prediction = "clear"
    if labels.count("blocked") > labels.count("clear"):
        prediction = "blocked"
    return [row[1] for row in near], prediction
for k in [1, 3]:
    print("query3", k, predict(3, k))
validation = [(3, "blocked"), (9, "blocked"), (11, "clear")]
training_labels = [row[2] for row in train]
baseline = "clear"
if training_labels.count("blocked") > training_labels.count("clear"):
    baseline = "blocked"
correct = baseline_correct = 0
for query, reference in validation:
    ids, prediction = predict(query, 3)
    print(query, ids, prediction, reference)
    correct += prediction == reference
    baseline_correct += baseline == reference
```

```
print("validation", correct, len(validation))
print("baseline", baseline, baseline_correct, len(validation))
```

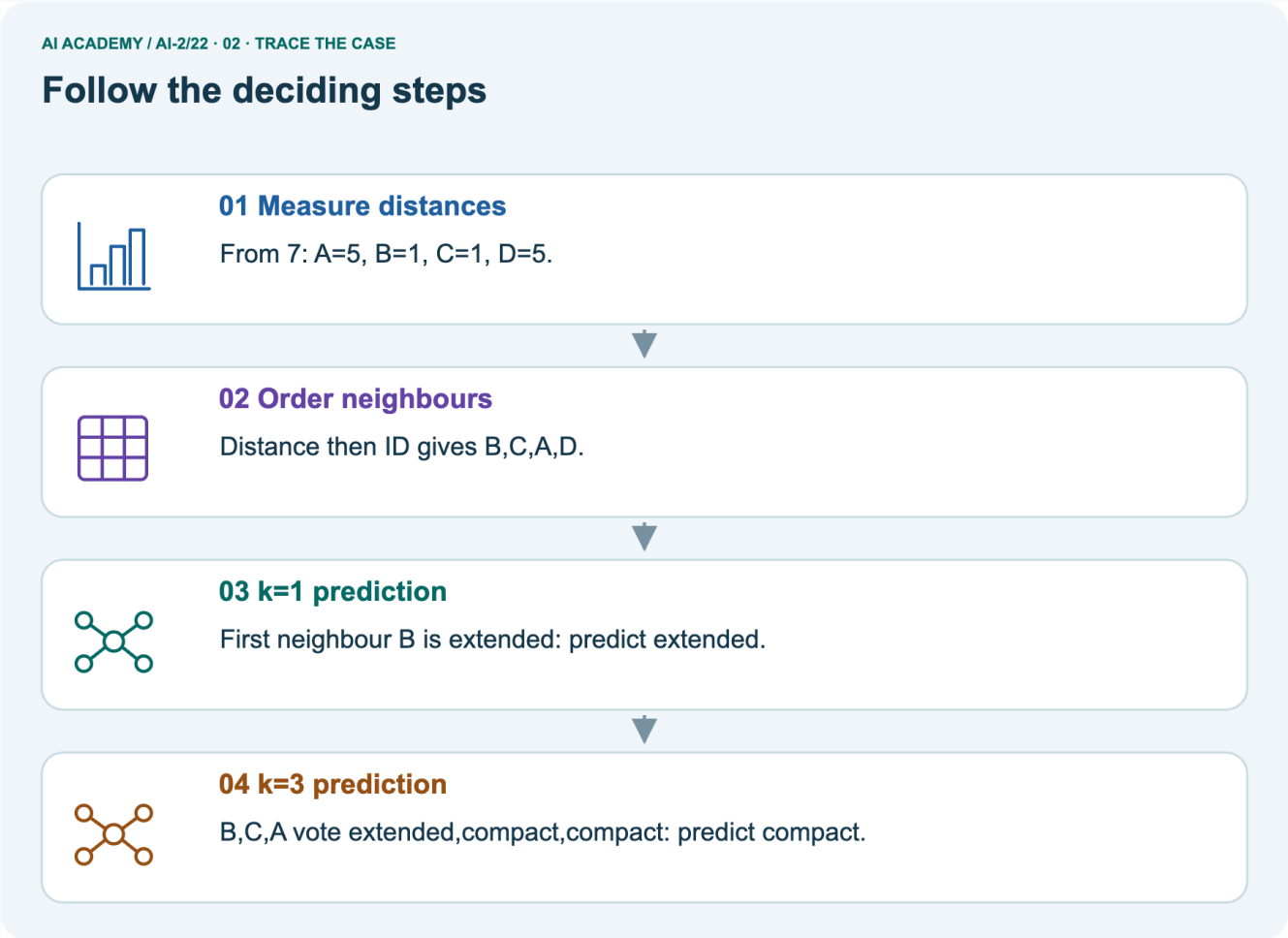
Read the separate four-book pool

A fictional one-feature classifier predicts compact or extended book format.

Stored **training** rows: A at 2 units compact, B at 6 extended, C at 8 compact, D at 12 extended. Query is 7.

Distance is absolute difference; ties sort by ID; tied category **vote** chooses compact.

Trace the book-query order and k choices



Follow the labelled picture with the matching explanation. Open the vector diagram

Changing k changes how many **training neighbour**s **vote**, even though all **distance**s remain fixed. The query's reference label must not enter the stored pool before predicting.

Tie policies make ordering reproducible; without them, list order could unexpectedly determine the single-**neighbour** result.

Nearest-**neighbour vote**s depend on feature representation; they are not calibrated probabilities or guaranteed correct labels.

Move only the book query

Change: Move query from 7 to 9 while keeping **training** fixed.

Prediction: Order becomes C,B,D,A; k=1 predicts compact and k=3 predicts extended.

Reason: **Distance**s are C=1,B=3,D=3,A=7; B beats D by the ID **tie rule**.

Elicit the picture explanation

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. New book's 3 nearest labels: adventure, adventure, mystery.

2. With k=3, adventure wins 2 **vote**s to 1.

3. With k=1, the closest single book controls the result.

4. Choose k and **tie rule** before viewing the desired answer.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

A=[1,1] liked, B=[1,0] disliked, C=[0,0] disliked; query=[0,1].

Find nearest **neighbour**s using differing-trait **distance** and k=1.

Exact explanation to model

A and C tie at **distance** one; B is **distance** two. Conflicting nearest labels require a predefined tie policy or uncertainty.

Explain the added worked example

Key terms

Term	Meaning
Distance	A numerical measure of difference between selected feature values
Neighbour	A stored training example close under the chosen distance
k	The number of neighbours selected for a prediction
Vote	A contribution of a selected neighbour's label to the predicted class

Model the reasoning aloud

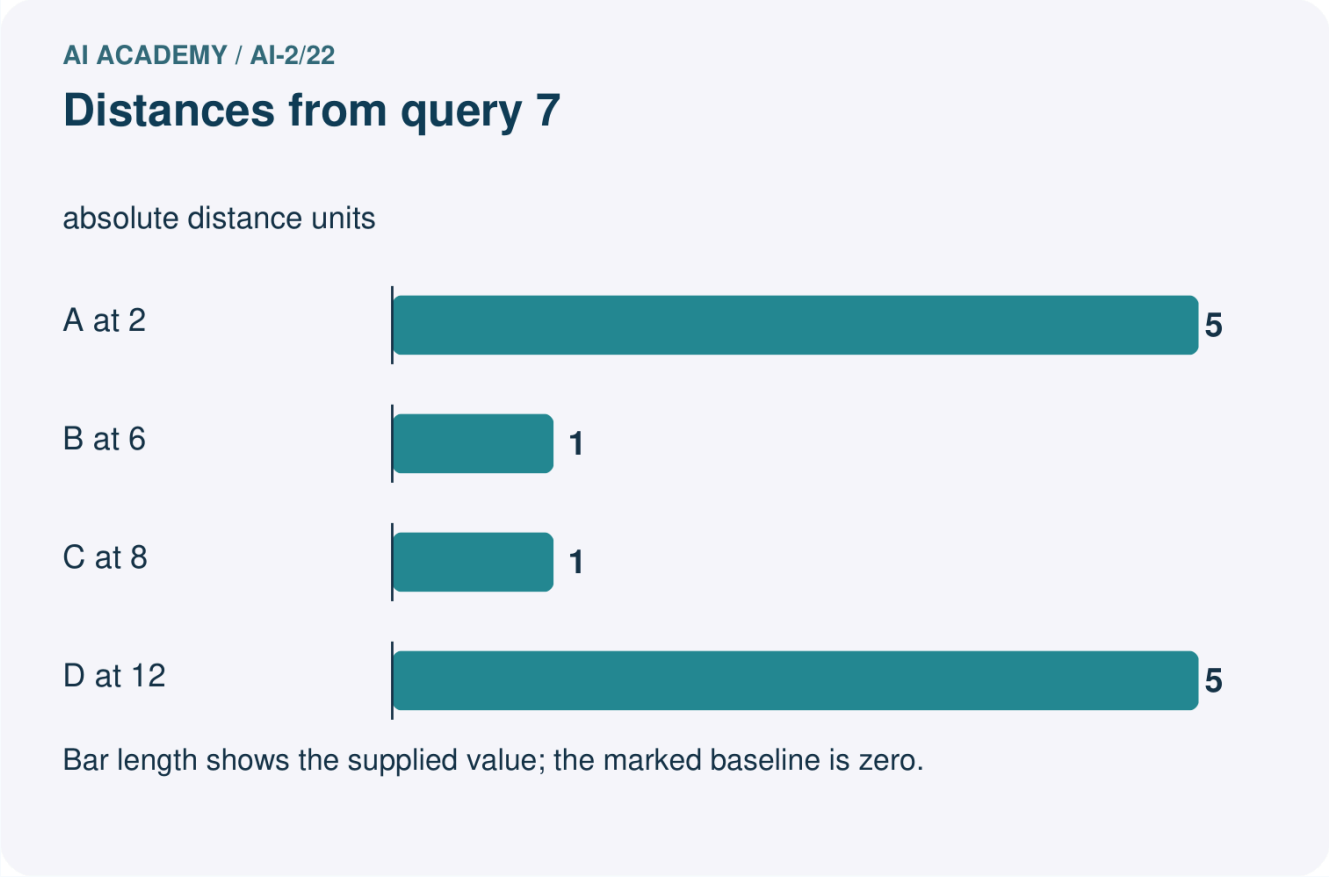
Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

A pictorial worked case: Nearest Neighbor Reasoning

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/22

Nearest Neighbor Reasoning

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Measure distances

From 7: A=5, B=1, C=1, D=5.



2. Order neighbours

Distance then ID gives B,C,A,D.



3. k=1 prediction

First neighbour B is extended: predict extended.



4. k=3 prediction

B,C,A vote extended,compact,compact: predict compact.

CHANGE ONE THING

Move query from 7 to 9 while keeping training fixed.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Measure distances	From 7: A=5, B=1, C=1, D=5.
2. Order neighbours	Distance then ID gives B,C,A,D.
3. k=1 prediction	First neighbour B is extended: predict extended.
4. k=3 prediction	B,C,A vote extended,compact,compact: predict compact.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the reasoning, then remove help

> Collect a response from each learner before discussing the worked key.

Prompt before revealing

Completion task: For query 3, list the selected IDs and **prediction**s for k=1 and k=3 using alphabetical ID order on equal **distance**s. Explain what changes and what remains the same.

Expected reasoning and fading help

Teacher check: For k=1 select A and predict blocked. For k=3 select A,B,C and predict clear from two clear **vote**s. **Training** data, **distance** and ID **tie rule** stay the same; the number of voting **neighbour**s changes.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Check the misconception before discussion

Use the worked case for Nearest **neighbour**s and **distance**s. Proposed explanation: "A negative difference is a negative **distance**". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Take the absolute difference; in this example **distance** is nonnegative regardless of which reading is larger.

Collect explanations, not only yes/no **vote**s. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Use the pictorial check

Why do k=1 and k=3 disagree for the original query?

The closest tied **neighbour** selected by ID is extended B, while the first three include two compact **vote**s.

Changed-case reasoning: Order becomes C,B,D,A; k=1 predicts compact and k=3 predicts extended. **Distance**s are C=1,B=3,D=3,A=7; B beats D by the ID **tie rule**.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: A=[1,1] liked, B=[1,0] disliked, C=[0,0] disliked; query=[0,1].

Write the question to explain: Find nearest **neighbour**s using differing-trait **distance** and k=1.

Reveal after **prediction**s: A and C tie at **distance** one; B is **distance** two. Conflicting nearest labels require a predefined tie policy or uncertainty.

Use the earlier changed question as optional later practice. Reserve Find **neighbour**s independently for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided: abs(2-3) and abs(4-3) both equal 1. For k=1 alphabetical tie order chooses A, blocked. k=3 adds B and C, whose clear **vote**s win 2 to 1. ID order is a deterministic convention, not **evidence** of the query's true label.

Exit: using labelled validation examples in the **neighbour** pool leaks evaluation answers and may let a query select itself. Restore the original **training**-only pool and score every validation query without adding its label.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve tree storage and prediction. Introduce prediction from stored examples with a declared distance.
7 to 18 minutes	Place A through E on the number line. Calculate absolute distances to query 3 and complete W1.
18 to 32 minutes	Trace tuple sorting and list slicing. Complete W2; distinguish tied distances from a tied label vote.
32 to 45 minutes	Read the diagram and answer the guided question. Explain why ID order is not evidence of truth.
Session 2 start to 8 minutes	Retrieve the training-only neighbour pool. Ask why validation labels must remain outside it.
8 to 20 minutes	Compute the three validation predictions and complete W3. Compare the fixed training-majority baseline.
20 to 35 minutes	Collect W4 independently before support. Require ordered IDs, both k predictions and the full validation score.
35 to 45 minutes	Use W5 and the exit question. Offer the fresh retry after feedback and record support.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

Query3 distances1,1,5,7,9; k1 selectsA blocked, k3 selectsA/B/C clear. Validation triples areA/B/C,C/D/E,D/E/C; all predictclear, matching only the third reference:1/3. Training-majority clear also1/3. The separate book query7 givesB/C/A/D, k1extended,k3compact; query9 givesC/B/D/A,k1compact,k3extended. These pools and labels are different. Never add query references before prediction, and do not call2/3 votes a calibrated probability. Reserve P–T for independent W4.

Run and record

1. Predict → Test → Record → Explain
2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.
3. Use the supplied five-row training pool, query3 and the three validation records. Record **distance**s, selected IDs, k, **vote**s and predefined ties; compare the frozen k3 method and training-majority baseline without adding query answers to training.

4. What did the hardest test teach you that the normal test did not?

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

A negative difference is a negative **distance**: Take the absolute difference; in this example **distance** is nonnegative regardless of which reading is larger.

Odd k prevents every kind of tie: Three binary **vote**s avoid a label tie, but **neighbour**s can still tie in **distance**, including at the selection boundary.

An exact match proves the label: It matches the selected feature only. Other conditions, annotation errors or contradictory examples can remain.

Choose support from the response

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Find neighbours independently for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected new case

> Keep this key until after the first attempt; accept valid alternatives with defensible reasoning.

Selected independent question and key

Selected case: New **training** rows are P at 1 cm clear, Q at 5 blocked, R at 7 blocked, S at 11 clear, T at 13 blocked.

For query 6, give all **distance**s, order the IDs by **distance** then alphabetically, and predict with k=1 and k=3. Using the same **training** pool and k=3, score validation (6,blocked), (10,clear),(12,blocked).

Compare the **training**-majority baseline.

Query 6 **distance**s P,Q,R,S,T are 5,1,1,5,7. Order Q,R,P,S,T. k=1 predicts blocked from Q; k=3 uses Q,R,P and also predicts blocked. For queries 6,10,12, selected triples are Q R P, S R T, S T R.

All predict blocked, matching validation rows 1 and 3: 2/3. **Training**-majority blocked also scores 2/3.

Judge explanation, trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Calculate feature distances

For query 3 cm, calculate **distance**s to all five **training** rows. Explain why A and B tie even though their labels differ.

Distances to A,B,C,D,E are 1,1,5,7,9 cm. A and B are equally far from 3 on the numeric feature; their labels do not enter the **distance** calculation.

W2 Apply neighbour and tie rules

For query 3, list the selected IDs and **prediction**s for k=1 and k=3 using alphabetical ID order on equal **distance**s. Explain what changes and what remains the same.

For k=1 select A and predict blocked. For k=3 select A,B,C and predict clear from two clear **vote**s. **Training** data, **distance** and ID **tie rule** stay the same; the number of voting **neighbour**s changes.

W3 Keep evaluation examples separate

Why must the three validation reference labels not enter the **neighbour** pool? Give the fixed k=3 score and baseline score on the worked validation examples.

Adding validation labels uses evaluation answers to make **prediction**s and changes the intended comparison with unseen trips. The fixed k=3 method predicts clear three times and gets 1/3.

The **training**-majority clear baseline also gets 1/3.

W5 Interpret a vote and a match

Two of three **neighbour**s **vote** clear. Does that prove a two-thirds chance of a correct **prediction**? Does an exact feature match guarantee a correct label? Explain both limits.

The **vote** fraction describes the selected **training** labels, not a verified probability of correctness. An exact match on one feature can still differ in omitted conditions or have a wrong or conflicting label.

Separate evaluation and relevant features remain necessary.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

A and C tie at **distance** one; B is **distance** two. Conflicting nearest labels require a predefined tie policy or uncertainty.

Additional transfer practice question

Training items have one numeric feature: A=2 with label short, B=8 with label long. A new item has value 6. Use absolute **distance** to choose its nearest **neighbour**.

Reasoned practice answer

Distances are 4 to A and 2 to B, so nearest-**neighbour prediction** is long. This follows the supplied **distance** rule, not certainty about every future case.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “The first stored example is always the nearest.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Row-order choice makes **prediction**s change when data is reordered. Predefine review for conflicting nearest labels, or a justified stable tie-breaker with disclosed limits.

Do not select a tie policy after looking at the correct answer for the test case.

Explanations for the additional workbook examples

W1 Calculate feature distances

W2 Apply neighbour and tie rules

W3 Keep evaluation examples separate

W4 Find neighbours independently

New **training** rows are P at 1 cm clear, Q at 5 blocked, R at 7 blocked, S at 11 clear, T at 13 blocked. For query 6, give all **distance**s, order the IDs by **distance** then alphabetically, and predict with k=1 and k=3.

Using the same **training** pool and k=3, score validation (6,blocked),(10,clear),(12,blocked). Compare the **training**-majority baseline.

Query 6 **distance**s P,Q,R,S,T are 5,1,1,5,7. Order Q,R,P,S,T. k=1 predicts blocked from Q; k=3 uses Q,R,P and also predicts blocked. For queries 6,10,12, selected triples are Q R P, S R T, S T R.

All predict blocked, matching validation rows 1 and 3: 2/3. **Training**-majority blocked also scores 2/3.

W5 Interpret a vote and a match

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

Use **neighbor**ing liked books as **evidence**, while explaining which traits created closeness.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Retain recommender **neighbour** IDs, **distance**s, k and tie decisions so similar-book explanations can be audited.

Support while preserving the reasoning goal

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Use physical steps on a number line before abs. Supply an empty **distance** table without the order.

Extension: explain how renaming tied IDs can change a **prediction** under this policy while the measured features remain unchanged, and why the policy must be recorded.

Require W4 **distance**s 5,1,1,5,7, order Q,R,P,S,T, both **prediction**s blocked, and validation plus baseline scores 2/3. Fresh retry: L at 0 blocked, M at 4 clear, N at 6 clear, O at 10 blocked, P at 12 clear; query 5.

Distances are 5,1,1,5,7; order M,N,L,O,P. k=1 is clear from M; k=3 is clear from M,N,L. Ask why L wins the third-place **distance** tie over O: alphabetical ID policy. Require a **training**-only pool and no probability claim.

Plan the delayed check

Suggested return points in this level: ai-2/23, ai-2/25, ai-2/28. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	

Record	Teacher evidence
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 23: Recommendation Scores. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week23: Recommendation Scores. Carry forward feature definitions, **training**-only evidence, **distance** and tie policies when contrasting a nearest-**neighbour** prediction with a declared recommendation score.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Transparent Book Recommender

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Compute absolute **distance**s and rank **training neighbour**s

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Practice and assessment reminders

Before class, solve the supported practice: For query 3, list the selected IDs and **prediction**s for k=1 and k=3 using alphabetical ID order on equal **distance**s. Explain what changes and what remains the same.

Reasoned practice answer: For k=1 select A and predict blocked. For k=3 select A,B,C and predict clear from two clear **vote**s. **Training** data, **distance** and ID **tie rule** stay the same; the number of voting **neighbour**s changes.

Check every learner before discussion: Use the worked case for Nearest **neighbour**s and **distance**s. Proposed explanation: “A negative difference is a negative **distance**”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Take the absolute difference; in this example **distance** is nonnegative regardless of which reading is larger.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Move query from 7 to 9 while keeping **training** fixed. Require a **prediction** and a reason before comparing. Expected result: Order becomes C,B,D,A; k=1 predicts compact and k=3 predicts extended.

Reason: **Distance**s are C=1,B=3,D=3,A=7; B beats D by the ID **tie rule**.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

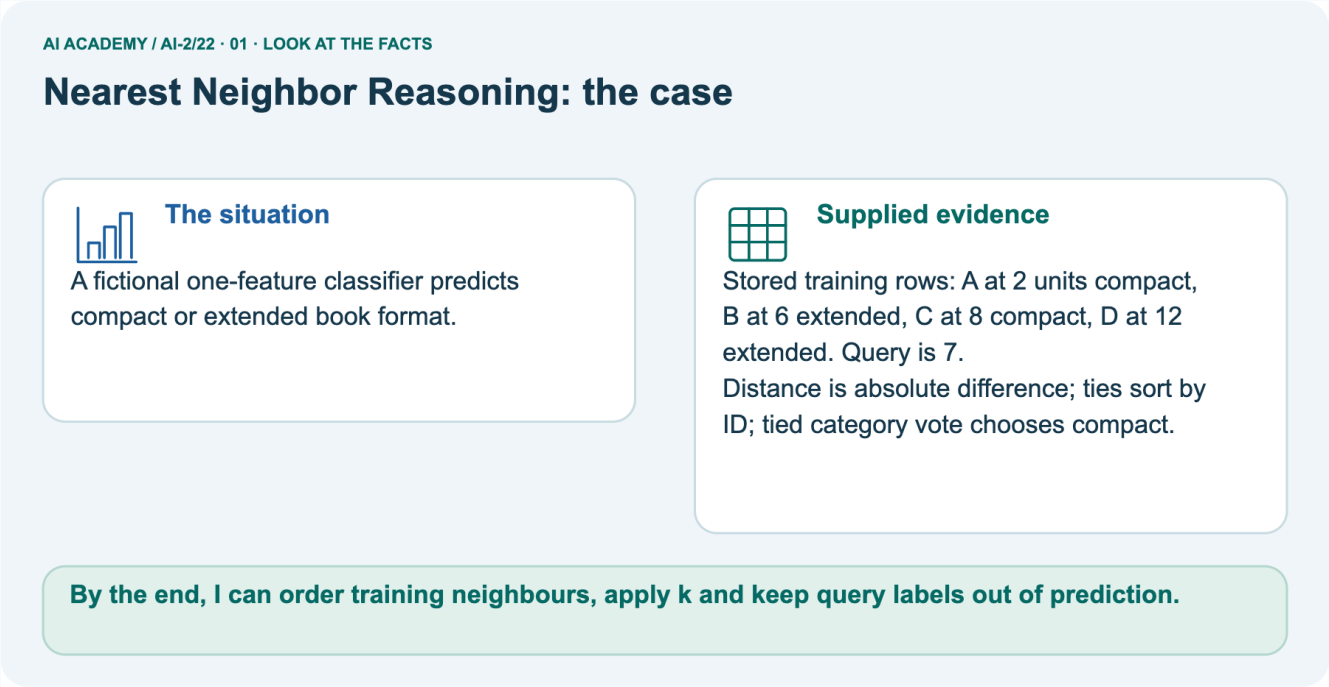
<details>
<summary>Reference: teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Nearest Neighbor Reasoning: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

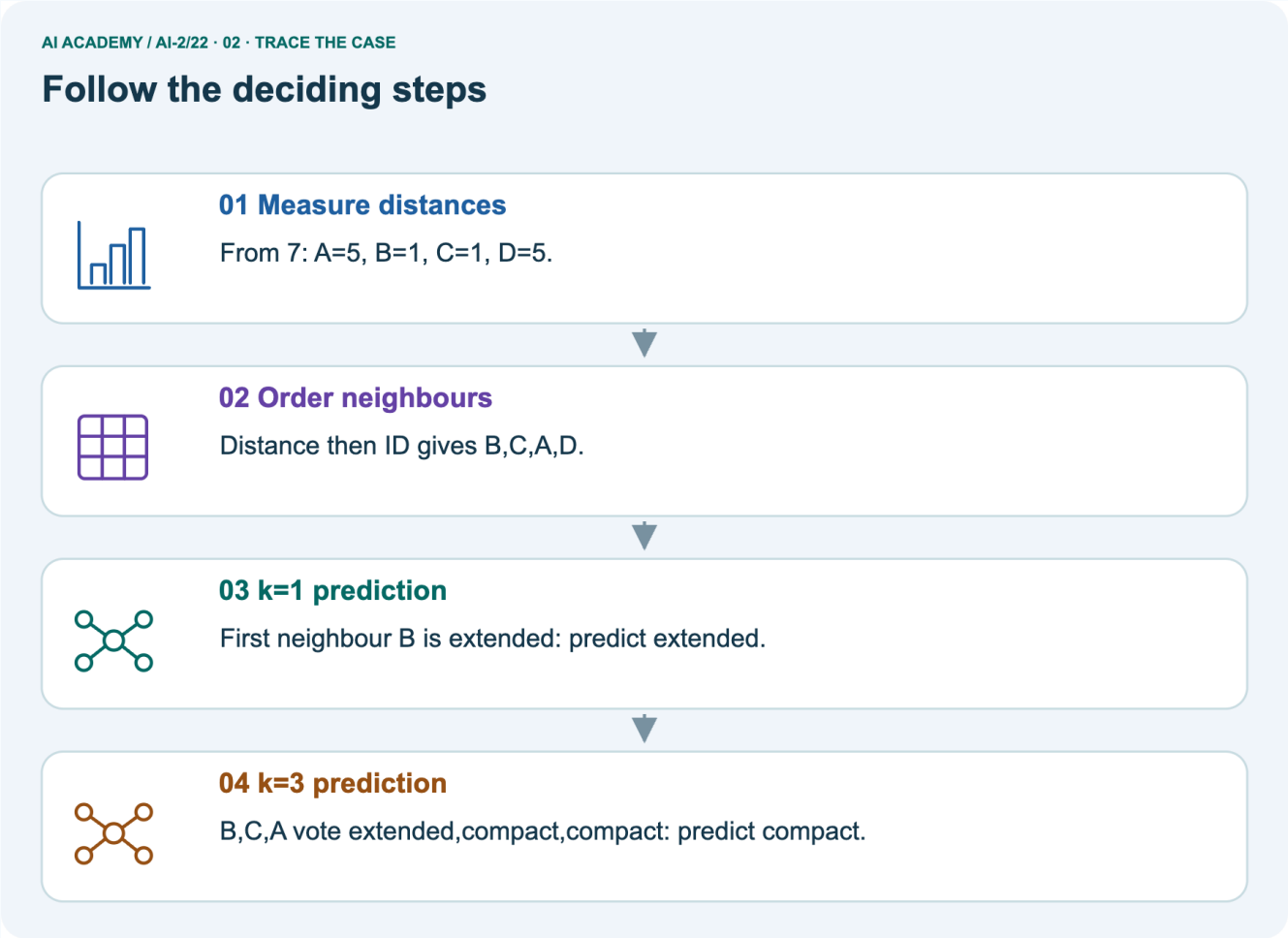
A fictional one-feature classifier predicts compact or extended book format. Supplied facts: Stored **training** rows: A at 2 units compact, B at 6 extended, C at 8 compact, D at 12 extended. Query is 7.

Distance is absolute difference; ties sort by ID; tied category **vote** chooses compact.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



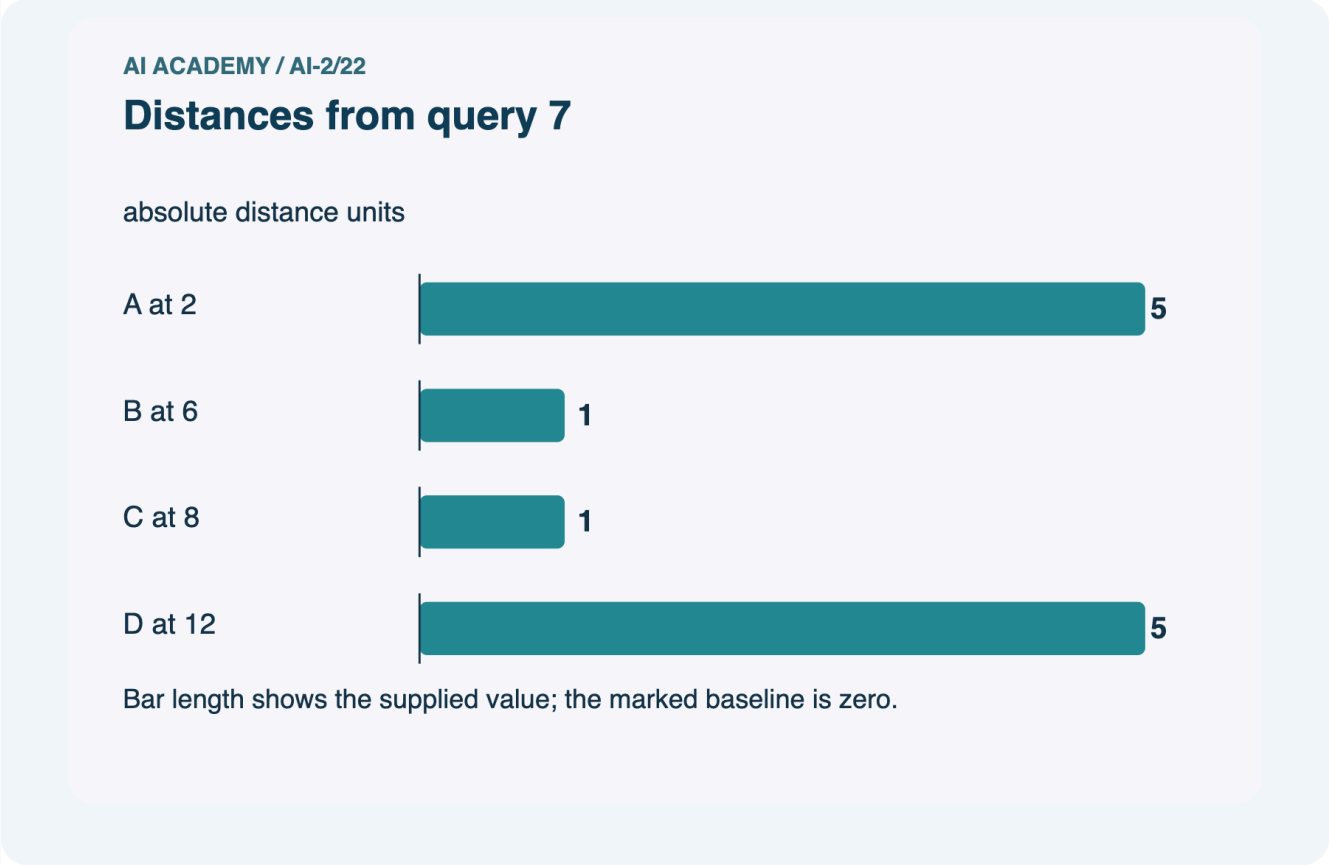
Follow the labelled picture with the matching explanation. Open the vector diagram

Measure **distance**s: From 7: A=5, B=1, C=1, D=5. → Order **neighbour**s: **Distance** then ID gives B,C,A,D. → k=1 **prediction**: First **neighbour** B is extended: predict extended. → k=3 **prediction**: B,C,A **vote** extended,compact,compact: predict compact.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Distances from query 7



Follow the labelled picture with the matching explanation. Open the vector diagram

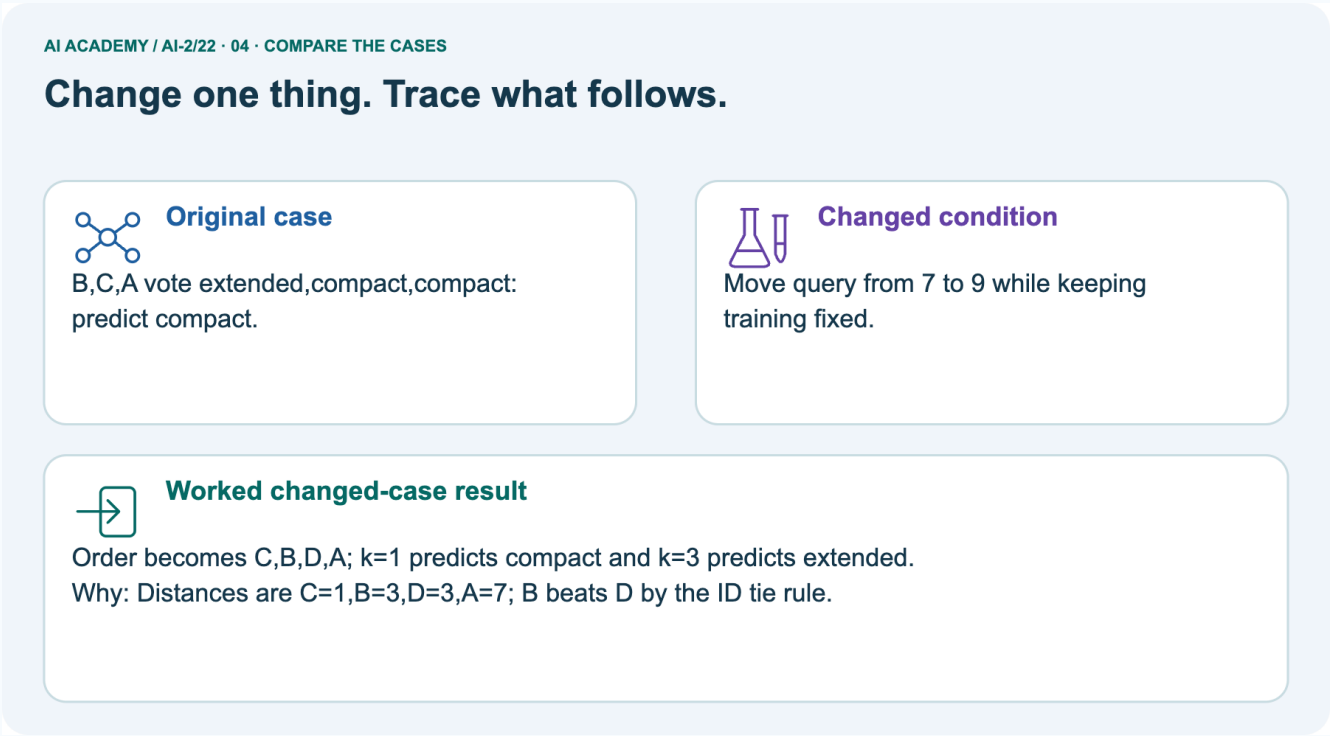
Distances from query 7. Units: absolute **distance** units. A at 2: 5. B at 6: 1. C at 8: 1. D at 12: 5.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: B,C,A **vote** extended,compact,compact: predict compact. Changed condition: Move query from 7 to 9 while keeping **training** fixed.

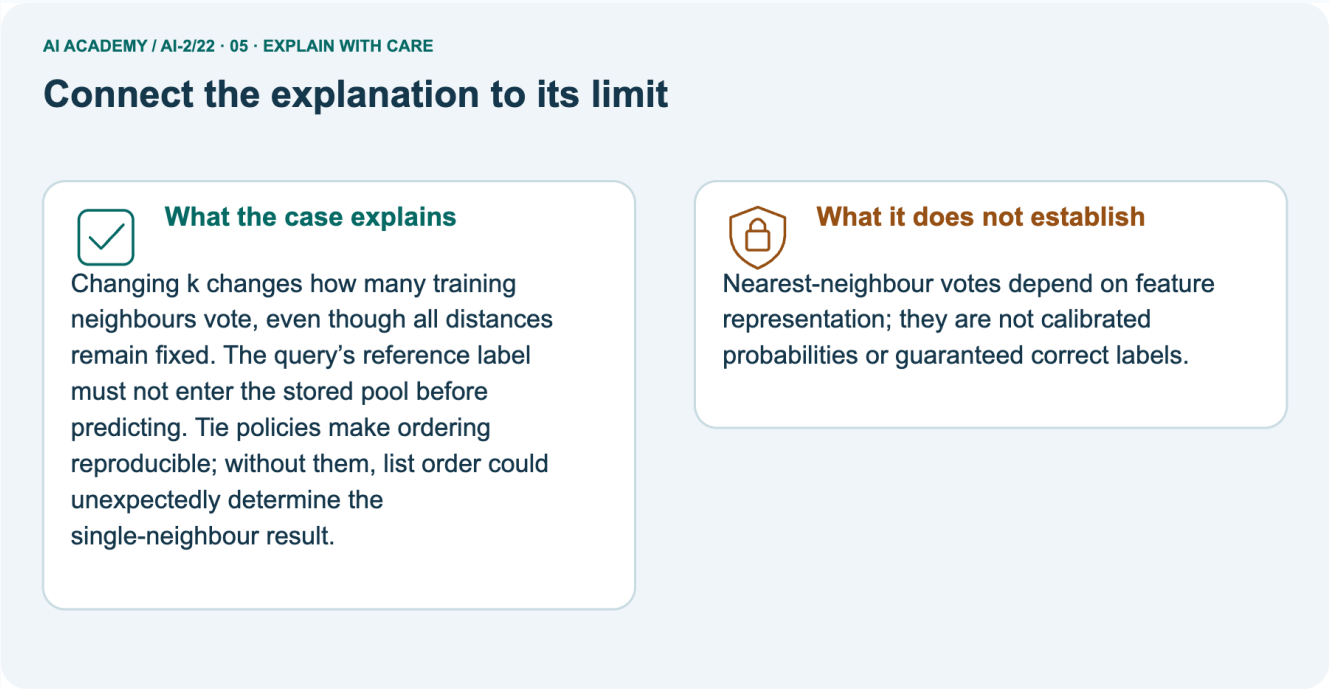
Expected result: Order becomes C,B,D,A; k=1 predicts compact and k=3 predicts extended. Reason: **Distance**s are C=1,B=3,D=3,A=7; B beats D by the ID **tie rule**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: **Distance**s are C=1,B=3,D=3,A=7; B beats D by the ID **tie rule**.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Changing k changes how many **training neighbour**s **vote**, even though all **distance**s remain fixed. The query's reference label must not enter the stored pool before predicting.

Tie policies make ordering reproducible; without them, list order could unexpectedly determine the single-**neighbour** result.

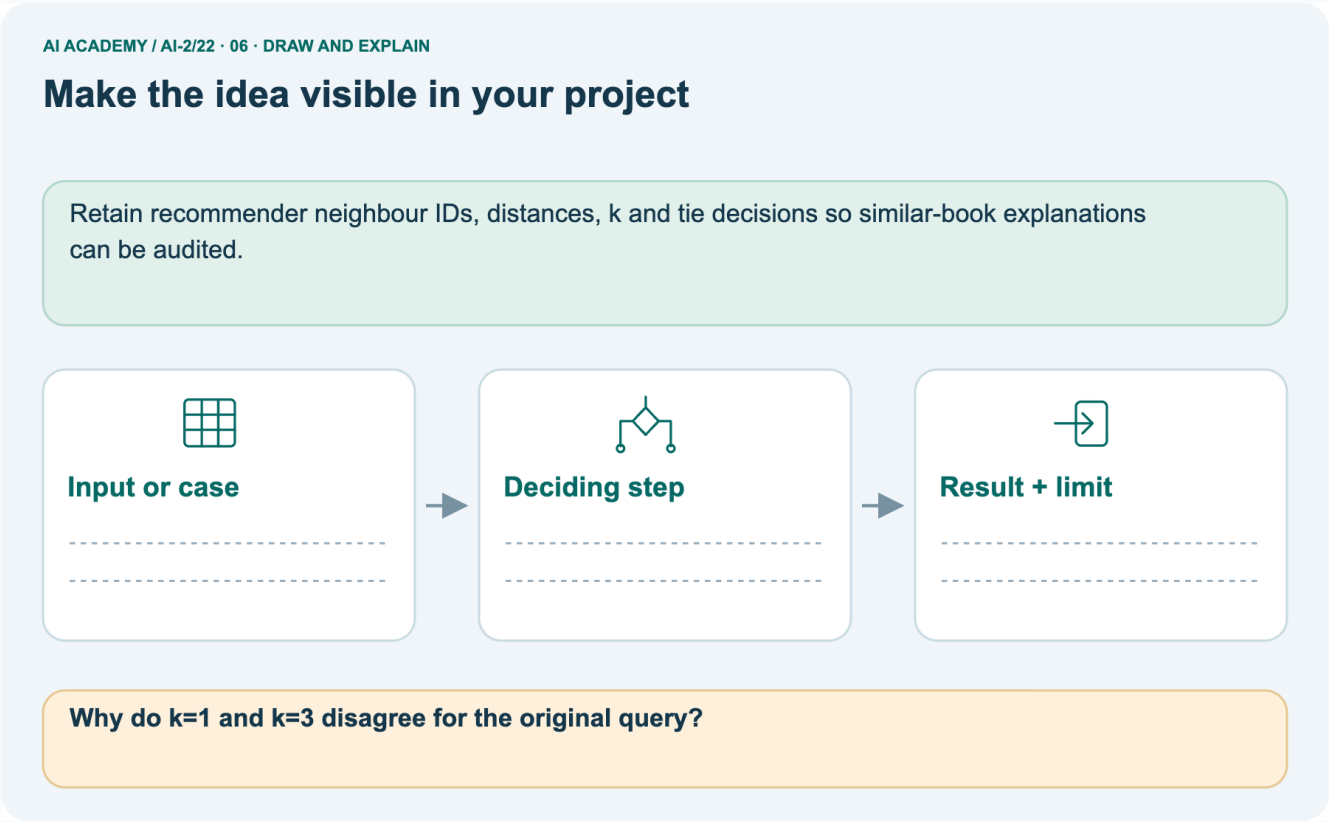
Limit: Nearest-**neighbour vote**s depend on feature representation; they are not calibrated probabilities or guaranteed correct labels.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Nearest-**neighbour vote**s depend on feature representation; they are not calibrated probabilities or guaranteed correct labels.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Retain recommender **neighbour** IDs, **distance**s, k and tie decisions so similar-book explanations can be audited. Draw your **input**, deciding step, result and limit.

Supported question: Why do k=1 and k=3 disagree for the original query?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The closest tied **neighbour** selected by ID is extended B, while the first three include two compact **vote**s.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Nearest Neighbor Reasoning: the case

02. Follow the deciding steps

03. Distances from query 7

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.